

BME 140 W24

Lab #4 Report

Lab: B1-2

AC Circuits and Filters

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Due: 2/22/24

Introduction

The purpose of this lab is to introduce AC circuits and filters. This lab had 4 objectives, to build a low and high pass filter, combine them to make a bandpass filter using the op-amp, test the frequency response of each filter, and finally compare the performance of the RC filter with an active filter. We will see how each filter attenuates its respective frequencies, low pass attenuating high frequencies, high pass attenuating low frequencies, and bandpass attenuating both high and low to allow a limited range of frequencies. By developing these circuits, the lab shows different types of voltage flow, and how that affects voltage gain.

Theoretical Background and Design

This lab utilizes the operational amplifier (op-amp). An op-amp is a voltage voltage-gaining device that can manipulate electrical signals by adding, multiplying, and even integrating. Five assumptions can be made for an op-amp: open loop gain is infinite and $\Delta V = 0$, no current flows through the input leads, there is no offset voltage or current, input impedance is infinite, and output impedance is equal to zero. When constructing circuits, there are four main concepts involved in this lab: the low pass RC filter, high pass RC filter, band passes RC filter, and bandpass active filter.

A) Low pass RC filter

The low pass filter is such that function generator input travels through a resistor then through a capacitor and ultimately to ground. To build an RC low pass filter with a cut-off frequency of 10kHz and an output impedance of 100k Ω in the fashion of the following figure:

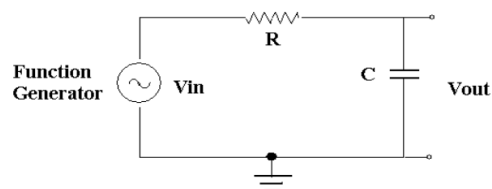


Figure 1: Low Pass RC Filter Diagram

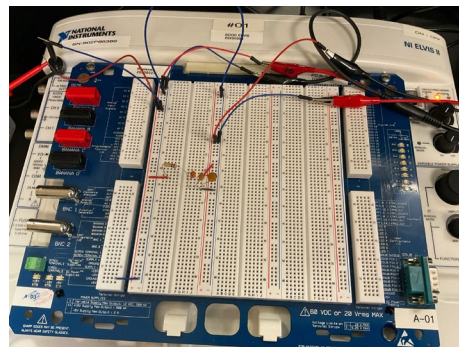


Figure 2: Low Pass RC Filter Wiring

We calculate what capacitor and resistor are needed for the low-pass RC circuit using the following formulas:

$$R = \frac{1}{10} Z_{out}$$

$$C = \frac{1}{2\pi R F_c}$$

We find $R = 100 \text{ k}\Omega$ (measured $9.89 \pm 0.01 \text{ k}\Omega$ using DMM) and $C = 1.59 \times 10^{-9} \text{ F}$. To compensate for the limited capacitors in the lab we chose 3 with values that will add up to our needed C when placed in parallel (820 pF, 680 pF, and 90 pF).

B) High Pass Filter

For the high pass filter, the function generator input travels to the capacitor first then the resistor, and then to the ground. To build an RC high pass filter (with a cutoff frequency of 100Hz and output impedance of 100k Ω), the diagram is as follows:

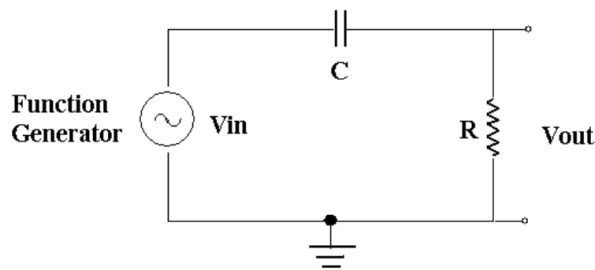


Figure 3: High Pass RC Filter

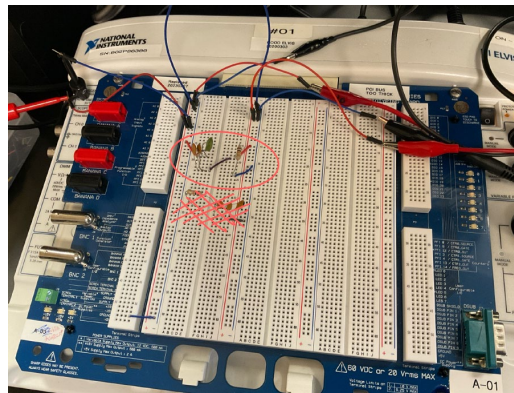


Figure 4: High Pass RC Filter Wiring

We repeat our previous calculation from the low pass filter and find $R=10 \text{ k}\Omega$ (measured at $9.89 \pm 0.01 \text{ k}\Omega$ using DMM) and $C=1.59 \times 10^{-12} \text{ F}$, again using 3 capacitors in parallel (0.1 μF , 0.05 μF , and 0.01 μF) to equal our needed capacitance.

C) Band Pass RC Filter

For the high pass filter, the function generator input travels to the capacitor first then the resistor, and then to the ground. To build a bandpass RC filter, the process is connecting the high pass and low pass filters in series to form an RC bandpass filter with cutoff frequencies of 100 Hz and 10 kHz like the below figure. Since we are just connecting the previous filters in series there is no change in the resistor or capacitor values.

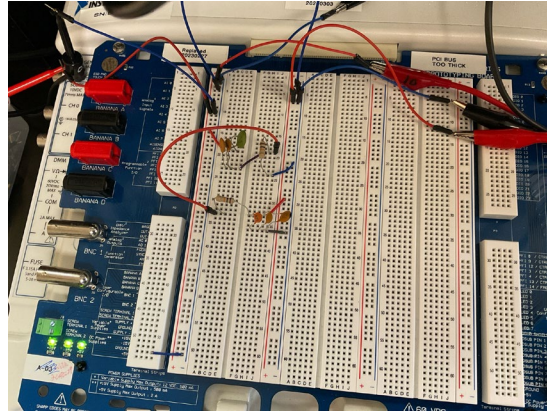


Figure 5: Band Pass RC Filter Wiring

D) Band Pass Active Filter

For the bandpass active filter, voltage input goes through the high pass filter. To build a bandpass active filter, 2 LM741 op-amps are used to build an active low pass and an active high pass filter. We then connect them in series to make an active band pass filter, the high pass being before the low pass. The active band has cutoff frequencies of 100 Hz and 10 kHz (and gain of 1) and capacitors $C_1 = 1.59 * 10^{-8}$ F (using a 150 pF and 10 pF in parallel to form an equivalent) and $C_2 = 1.59 * 10^{-13}$ F (using 0.0010 μ F and 0.015 μ f) respectively.

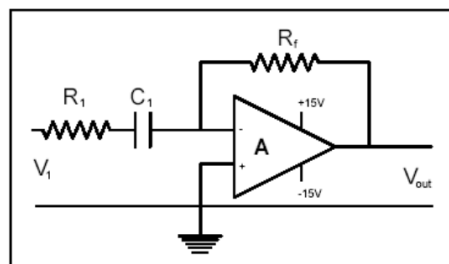


Figure 6: High Pass Active Filter

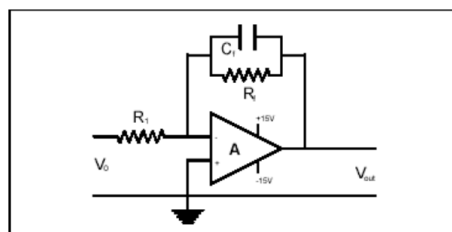


Figure 7: Low Pass Active Filter

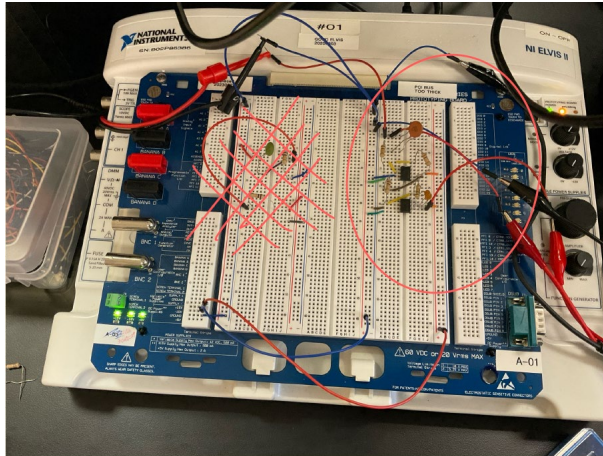


Figure 8: Band Pass Active Filter Wiring

Results and Discussions

A) Low Pass RC Filter

Frequency (Hz)	20log of frequency	Ain (V)	Aout (V)	Gain
100	40	4	4.396 ± 0.001	1.099
1000	60	4	4.396 ± 0.001	1.099
3000	69.54243	4	4.236 ± 0.001	1.059
10000	80	4	3.227 ± 0.008	0.80675
30000	89.54243	4	1.678 ± 0.020	0.4195
100000	100	4	0.4311 ± 0.010	0.107775
1000000	120	4	0.06 ± 0.005	0.015

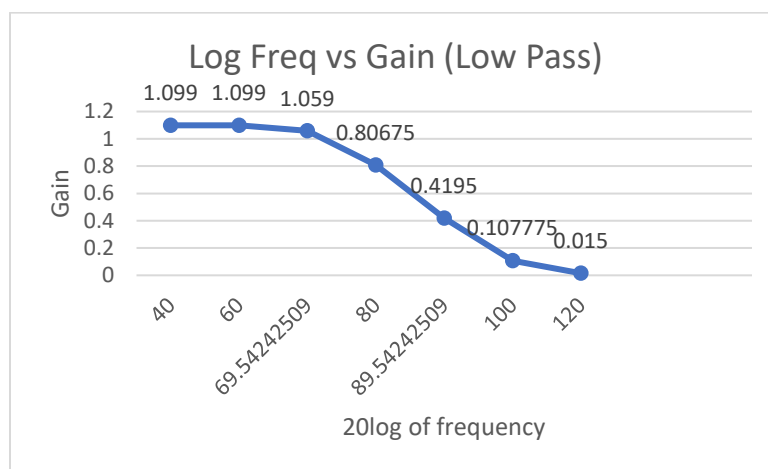


Figure 9: Graph of Gain Magnitude of Low Pass RC Filter

- From the main plot, does the simple RC circuit justify the name "low-pass filter"? Why?

Yes, as soon as the input exceeds 1kHz which we can consider high frequency, the output begins to attenuate, meaning that only low frequencies can pass through the filter.

- b. Find the frequency at which the gain magnitude, $G = A_{out}/A_{in}$, drops to $1/\sqrt{2} = 0.707$. This should be the cutoff frequency, f_c .

Seeing from our data that at the cutoff frequency of 10kHz, the Gain is .80675 and not 0.707, if data from 10 kHz to 30 kHz is linear, we can find the slope from the graph using the following slope equation:

$$\frac{0.80675 - 0.4195}{10,000 - (-10,000)} = 1.9 \times 10^{-5}$$

then plugging into the equation of a line:

$$0.707 = 1.9 \times 10^{-5} * F_{cl} + 1$$

We find the measured cutoff frequency to be 15421 Hz.

- c. Does f_c agree with the theoretical result?

Yes, as it is within a 25% range of 10 kHz and is close enough to still be considered a low pass.

B) High Pass RC Filter

Frequency (Hz)	20 log of frequency	A _{in} (V)	A _{out} (V)	Gain
1	0	4.197	0.2789 ± 0.0001	0.066452
10	20	4.197	0.6005 ± 0.03	0.143078
30	29.54243	4.197	1.314 ± 0.040	0.313081
100	40	4.197	2.908 ± 0.001	0.692876
300	49.54243	4.197	3.905 ± 0.030	0.930426
1000	60	4.197	4.184 ± 0.040	0.996903
10000	80	4.197	4.223 ± 0.001	1.006195

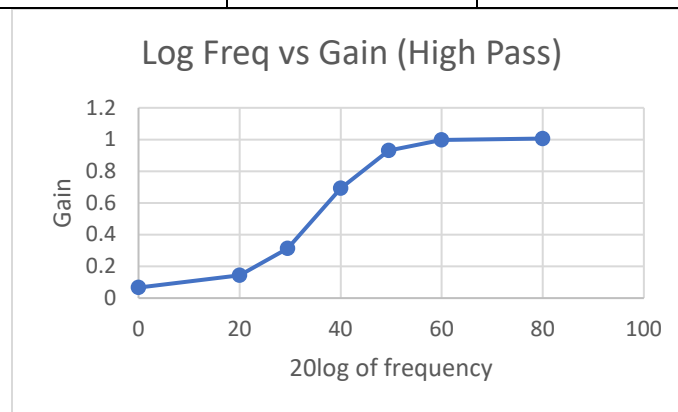


Figure 10: Graph of Gain Magnitude of High Pass RC Filter

- a. From the gain plot, does the simple RC circuit justify the name “high-pass filter”? Why?

Yes, most of the low frequencies are attenuated and all the high frequencies pass through the circuit.

- b. Find the frequency at which the gain magnitude, $G = A_{out}/A_{in}$, drops to $1/\sqrt{2} = 0.707$. This should be the cutoff frequency, f_{cH} .

Repeating the process from part A, our closest value to 0.707 is at 100 Hz with a gain value of 0.69. Assuming the data from 100 Hz to 300 Hz is linear, we can find the slope using the same equation.

$$\frac{0.930426 - 0.692876}{100 - (-100)} = 1.18 \times 10^{-3}$$

then plugging into the equation of a line:

$$0.707 = 1.18 \times 10^{-3} * F_{cL} + 1$$

We find the measured cutoff frequency to be 246 Hz.

- c. Does f_{cH} agree with the theoretical result?

This cutoff frequency value agrees with the cutoff frequency of 100 Hz as it is within reason when considering a spectrum that can range from 1 Hz to 10 kHz. This was perhaps due to mechanical issues in the circuit and variable capacitor/resistor values, through all capacitors and resistors were measured with the DMM.

C) Band Pass RC Filter

Frequency (Hz)	20 log of frequency	A _{in} (V)	A _{out} (V)	Gain
1	0	4.197	0.0711 ± 0.0008	0.016941
10	20	4.197	0.423 ± 0.004	0.100786
30	29.54243	4.197	1.179 ± 0.001	0.280915
100	40	4.197	2.738 ± 0.001	0.652371
300	49.54243	4.197	3.865 ± 0.001	0.920896
1000	60	4.197	4.064 ± 0.001	0.968311
3000	69.54243	4.197	3.905 ± 0.001	0.930426
10000	80	4.197	2.948 ± 0.001	0.702406
30000	89.54243	4.197	1.26 ± 0.010	0.300214
100000	100	4.197	0.4234 ± 0.001	0.100882
1000000	120	4.197	0.05 ± 0.005	0.011913

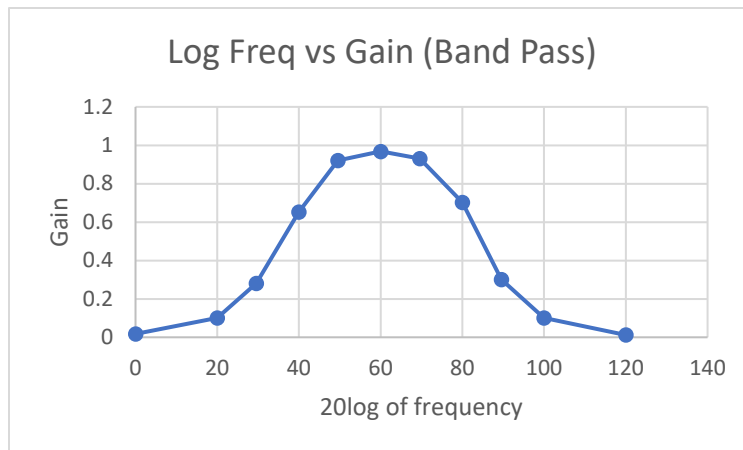


Figure 11: Graph of Gain Magnitude of Band Pass RC Filter

- a. From the gain plot, does the simple RC circuit justify the name “band-pass filter”? Why?

Yes, because the low and high frequencies are attenuated but this time there is a longer range in the middle of what can pass.

- b. Find the frequency at which the gain magnitude, $G = A_{out}/A_{in}$, drops to $1/\sqrt{2} = 0.707$. This should be the cutoff frequency, f_{cL} and f_{cH} .

The process repeats except now with two values for f_{cL} and f_{cH} . We use 100 Hz and 3000 Hz to assume linearity with the following frequency to assume linearity as those are our closest values within the range to find our values with a gain of 0.707.

$$\frac{0.9209 - 0.65237}{100 - (-100)} = 1.34 \times 10^{-3}$$

then plugging into the equation of a line:

$$0.707 = 1.34 \times 10^{-3} * F_{cH} + 1$$

Repeat the process for the other cutoff frequency:

$$\frac{0.93043 - 0.70241}{3000 - (-3000)} = 3.8 \times 10^{-5}$$

then plugging into the equation of a line:

$$0.707 = 3.8 \times 10^{-5} * F_{cL} + 1$$

Our cutoff frequencies are 218 Hz and 7710 Hz.

- c. Do f_{cL} and f_{cH} agree with the value measured in A and B? Why?

This agrees with the cutoff frequencies of 100 Hz and 3 kHz as it is within respectable ranges of 100 Hz and 3 kHz.

D) Band Pass Active Filter

Frequency (Hz)	20 log of frequency	Ain (V)	Aout (V)	Gain
1	0	4.197	0.07882 ± 0.005	0.01878
10	20	4.197	0.4023 ± 0.00001	0.095854
30	29.54243	4.197	1.07 ± 0.010	0.254944
100	40	4.197	2.738 ± 0.001	0.652371
300	49.54243	4.197	3.984 ± 0.010	0.949249
1000	60	4.197	4.233 ± 0.001	1.008578
3000	69.54243	4.197	4.064 ± 0.040	0.968311
10000	80	4.197	3.068 ± 0.049	0.730998
30000	89.54243	4.197	1.155 ± 0.001	0.275197
100000	100	4.197	0.31 ± 0.003	0.073862
1000000	120	4.197	0.015 ± 0.002	0.003574

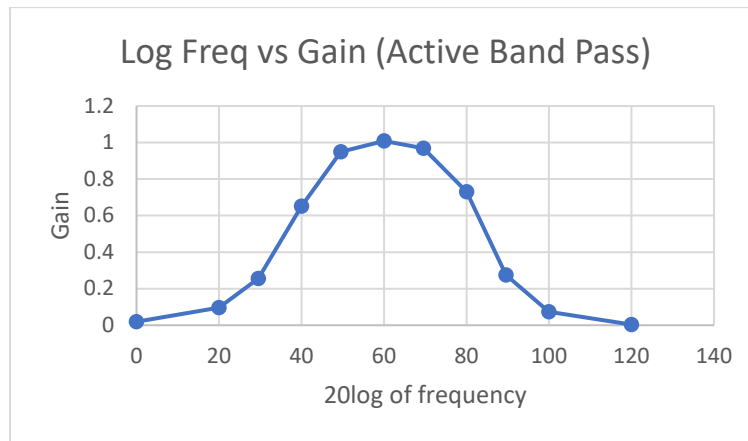


Figure 12: Graph of Gain Magnitude of Active Band Pass RC Filter

- a. From the gain plot, does the simple RC circuit justify the name “band-pass filter”? Why?

Yes, because the low and high frequencies are attenuated but this time there is a longer range in the middle of what can pass.

- b. Find the frequency at which the gain magnitude, $G = A_{out}/A_{in}$, drops to $1/\sqrt{2} = 0.707$. This should be the cutoff frequency, f_cL and f_cH .

The process repeats except now with two values. We use 100 and 3000 Hz to assume linearity with the following frequency to assume linearity to find our values with a gain of 0.707.

$$\frac{0.94925 - 0.65237}{100 - (-100)} = 1.48 * 10^{-3}$$

then plugging into the equation of a line:

$$0.707 = 1.48 * 10^{-3} * F_{cH} + 1$$

Repeat the process for the other cutoff frequency:

$$\frac{0.731 - 0.2752}{10000 - (10000)} = 2.279 * 10^{-5}$$

then plugging into the equation of a line:

$$0.707 = 2.279 * 10^{-5} * F_{cL} + 1$$

Our cutoff frequencies are 197 Hz and 12857 Hz.

c. Does f_{cL} and f_{cH} agree with the value predicted?

This agrees with the cutoff frequencies of 100 Hz and 10 kHz as it is within reasonable ranges of 100 Hz and 10 kHz.

d. Compare Active filters with RC filters, what is the advantage of active filters?

The advantage of an active filter over an RC filter is that it can handle the load, meaning that higher values of impedance that may hinder the function of an RC will not impact the active filter. Also, the op-amp is there to keep the input impedance high and the output impedance low allowing each filter to do what is required. This means there is no attenuation between filters.

Conclusion

Throughout the lab, we delved into constructing various types of filters including RC high-pass, low-pass, and band-pass filters, each tailored to attenuate specific frequency ranges. Additionally, we expanded our repertoire by integrating LM741 op-amp circuits to devise active high-pass, low-pass, and band-pass filters. We determined the frequencies at which attenuation occurs for each filter, elucidating the nuances of their performance. By manipulating resistor-capacitor relationships, such as employing a 10 k Ω resistor and 0.16 μ F capacitor to establish cutoff frequencies based on equation $f = 1/(2\pi RC)$, we honed our ability to design filters with precision. Combining these into an active filter allowed for a better understanding of having two cutoff frequencies for a single circuit as the high pass would have capacitor values increased by a factor of 10^x compared to that of the low pass filter. This comparative exploration highlighted the superior accuracy of active filters owing to their integration of op-amps, underscoring their efficacy in signal processing applications.

